USRI (UNDERGRADUATE SUMMER RESEARCH INTERNSHIP) RESEARCH OUTPUT SUMMER 2026

Using LLMs to Determine the Mechanisms of Missing Data

Treating missing data as a causal inference problem and testing if a Large Language Model (LLM) can identify the cause behind the gaps.

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01 CONTEXT AND MOTIVATION

A value is missing when a cell that should hold an observation is empty.

MedInc	HouseAge	AveBedrms	MedHouseVal
8.33	41	1.02	4.53
8.30	21	?	3.59
7.26	52	1.07	?

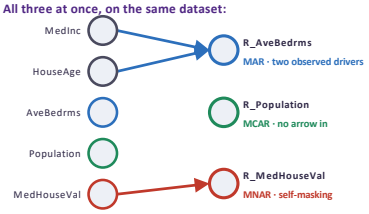
- Gaps are rarely random and often follow a pattern
- Approx. 96% of studies just delete incomplete rows
- To fill a blank, first find what removed it

02 BACKGROUND: M-GRAPHS

Rubin classifies a gap by what its indicator R depends on. Pearl draws it as an m-graph.

MCAR (Missing Completely At Random) P(R | X) = P(R)
MAR (Missing At Random) P(R | X) = P(R | Xobs)
MNAR (Missing Not At Random) P(R | X) depends on Xmis

All three at once, on the same dataset:



03 GOAL AND OBJECTIVES

- **Goal:** test if an LLM can propose the missingness graph from the broken data alone.

Dataset with holes + summary statistics
↓
LLM
↓
Proposed m-graph + probability tables
↓
Compare with the true m-graph

Objectives: find the affected columns, name each mechanism & recover accurate probabilities.

04 DATA AND AMPUTATION

California Housing: 20,640 census blocks from the 1990 US census, eight numeric predictors plus the median house value.

`MedInc · HouseAge · AveRooms · AveBedrms · Population · AveOccup · Latitude · Longitude · MedHouseVal`

Semi-synthetic data = real values + synthetic missingness whose graph and CPTs I keep.

- Bin numeric columns into quantiles for conditioning
- Write one CPT per mechanism: P(missing | parents)
- Delete cells by a seeded Bernoulli draw (seed 42)

05 THE PROMPT AND THE RESPONSE

Two separate calls. The solver never sees the CPTs, the mask or the true graph.

MISSINGNESS IMPUTATION PROMPT

```
Specify each mechanism as a conditional
probability table: P(missing | parents).
MNAR parents MUST include the target.
```

CAUSAL REASONING PARTNER PROMPT

```
You are a partner in causal reasoning:
understand WHY data is missing, not THAT.
Read > Generate > Challenge > Predict >
Commit, then a CPT per column, as JSON.
```

ONE HYPOTHESIS RETURNED

```
"affected_column": "AveBedrms",
"proposed_cause": "MedInc, HouseAge",
"suspected_mechanism": "MAR", "confidence": "high"
```

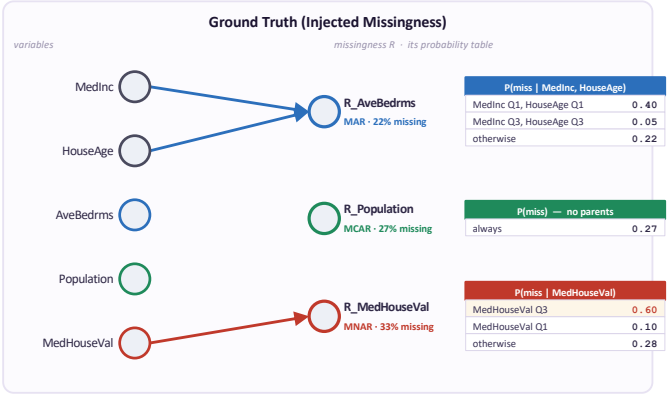
06 SCORING AND IMPLEMENTATION

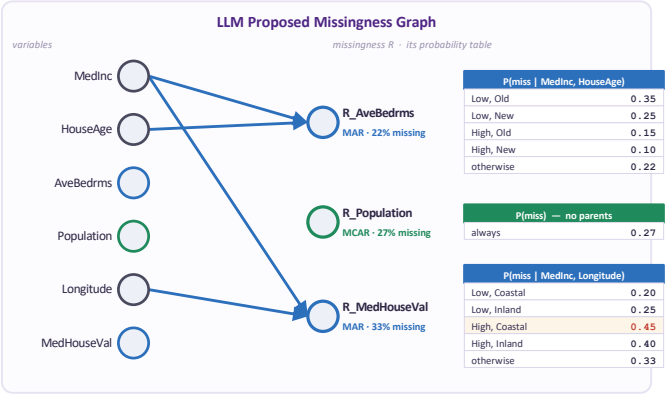
- Columns found, and columns invented
- Mechanism confusion matrix (MCAR / MAR / MNAR)
- True vs spurious edges recovered
- CPT error, cell by cell (MAE / RMSE)

Implementation: a dataset-agnostic CPT injector with seeded RNG, an automatic m-graph renderer, the elicitation prompt and the scoring harness.

07 EXPERIMENTAL RESULTS

One run on California Housing (seed 42), model Gemini-3.1-flash-lite. Each R node carries the CPT that drives it.





3 / 3 affected columns found · **0** columns invented · **2 / 3** mechanisms named correctly · **2 / 3** true edges recovered, 2 spurious generated

HOW DID THE LLM PERFORM?

- It often reads correlation as causation: MedInc and Longitude proxy house value, so the self-mask became two observed drivers.
- MNAR leaves no distinct trace in data alone; diagnosing it needs subject expertise the model does not have.
- Where the structure was right, so were the probabilities: Population exact, every AveBedrms cell within 0.05.

08 CONCLUSION, TAKEAWAYS AND FUTURE WORK

- **Capable, given accurate context.** Every affected column was found and no missingness was invented.
- **MNAR is a ceiling.** A self-mask is not identifiable from observable data; the model would need domain training.
- **Future work.** Compare models head-to-head; train a model in a given field.

References Mohan, Pearl & Tian (2013), Missing data as a causal inference problem, UCLA TR R-410 · Mohan & Pearl (2021), JASA 116(534) · Pearl (2009), Causality, 2nd ed. · Rubin (1976), Biometrika 63(3) · Little & Rubin (2002), Statistical Analysis with Missing Data · Peugh & Enders (2004), Rev. Educ. Res. 74(4) · California Housing: Pace & Barry (1997), Statistics & Probability Letters 33(3)